Little Lab Coats

🌿 What Is an Ecosystem?

Grade 2 Life Science, Unit 2 — Lesson 1

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Ecosystems 🌱 **Unit:** 2-LS2 Ecosystems

Living and nonliving things working together — that is an ecosystem!

📋 **NGSS Standard:**
2-LS2-1 — Make observations of plants and animals to compare the diversity of life in different habitats.

🎯 Learning Objectives

- Identify the living and nonliving parts of an ecosystem
- Understand that living things depend on their environment to survive
- Recognize that all parts of an ecosystem are connected

📦 Materials

🏠 What You Need

- Paper and pencil (for the Ecosystem Map)
- Green and blue crayons or colored pencils
- Window or outdoor access (a few minutes of looking outside)
- Magnifying glass — optional but fun!

✅ *No specialty purchases needed — paper and crayons only!*

📖 Vocabulary

Ecosystem
All the living and nonliving things in one place, working together

Living
Something that grows, breathes, and needs food and water

Nonliving
Something that was never alive (rocks, water, sunlight)

Habitat
The place where an animal or plant lives and finds what it needs

Environment
Everything that surrounds a living thing: air, water, soil, other living things

Depend
To need something else in order to survive

🌍 Background

An ecosystem is made up of all the living and nonliving things in one place. Trees, insects, birds, and soil bacteria are living parts. Rocks, sunlight, water, and air are nonliving parts. Everything in an ecosystem is connected — change one part and the whole ecosystem feels it.

Living things in an ecosystem depend on the nonliving parts to survive. Plants need sunlight and water to grow. Animals need clean air to breathe and fresh water to drink. When one part of an ecosystem changes (for example, if a drought stops the rain), all the living things are affected.

Ecosystems come in all sizes. A puddle can be a tiny ecosystem for bacteria and water insects. A rainforest is a giant ecosystem with millions of species. Even a backyard, a park, or a window box is an ecosystem you can explore today. No matter how small, every ecosystem has the same ingredients: living things, nonliving things, and the connections between them.

🔬 Activity: Ecosystem Map

In this activity, you will look carefully at the outdoor world around you and make a map of the ecosystem you can see.

Step 1: Look out a window or step outside for 5 minutes. Bring paper and pencil.

Step 2: Draw a simple map of what you can see — trees, sky, soil, grass, sidewalk, birds, insects, rocks, puddles, and sunlight. It does not need to be perfect — just get everything you notice onto the paper.

Step 3: Go back through your drawing and circle every **living thing** in green. Circle every **nonliving thing** in blue.

Step 4: Count how many living things you found and how many nonliving things. Write the numbers at the bottom of your map.

Step 5: Look at your map again. Draw a dotted arrow from one living thing to a nonliving thing it depends on. For example: draw an arrow from a tree to the sunlight, or from a bird to a tree.

Step 6: Share your map and discuss: "What surprised you about how many connections you found?"

💬 Discussion Questions

🗣️ **Question 1:** If the rain stopped falling for an entire year, which living things would be most affected first?

Suggested answer: Plants would be affected first because they need water to grow. Without plants, plant-eating animals would lose their food. Then meat-eating animals would lose their food too. The whole ecosystem would change, starting with the plants — they are the foundation of most ecosystems.

🗣️ **Question 2:** Is a rock a living or nonliving thing? How do you know?

Suggested answer: A rock is nonliving because it does not grow, breathe, eat, or reproduce. It was never alive. Soil is interesting though — it contains tiny living things called bacteria and fungi. The dirt itself is nonliving, but the organisms living inside it are alive. This shows that living and nonliving things are always mixed together.

🗣️ **Question 3:** Can a living thing survive without any nonliving things nearby?

Suggested answer: No! Every living thing needs nonliving things. Plants need sunlight (nonliving), water (nonliving), and minerals in soil (nonliving). Animals need air (nonliving) and water (nonliving). Life and nonlife are always working together in every ecosystem on Earth.

🌿 Student Worksheet — What Is an Ecosystem?

Name: _____ Date: _____

Part 1 — My Ecosystem Map

Draw everything you can see outside your window or yard. Circle living things in GREEN. Circle nonliving things in BLUE.

✏️ *Draw your ecosystem map here...*

Part 2 — What Did I Find?

🌿 Living Things I Found	💧 Nonliving Things I Found

Total living things: _____ Total nonliving things: _____

Part 3 — Fill in the Blanks

An ecosystem is made up of _____ and _____ things working together.

One living thing in my ecosystem: _____ depends on _____ to survive.

👩‍🏫 Parent/Teacher Notes

- **No materials to buy:** Paper and crayons only. A magnifying glass adds excitement if you have one — even a cheap plastic one transforms the experience.
- **If outdoor access is limited:** Look out a window, or describe the ecosystem of a houseplant (soil, sunlight from window, the plant itself, and any bugs you might find). Even a kitchen windowsill is an ecosystem.
- **Connection to prior learning:** Students may have learned about animal needs (food, water, shelter, air). This lesson zooms out to show HOW animals get those needs met — from their ecosystem.
- **Differentiation:** Younger learners can sort picture cards into "living" and "nonliving" piles. Advanced learners can research a specific ecosystem (desert, ocean, rainforest) and compare it to their backyard.
- **This lesson is the foundation for the whole unit.** Spend extra time on the idea that everything is connected. The Ecosystem Map activity makes that connection visible and personal.
- **Common misconception:** Students sometimes think soil is nonliving. Soil contains billions of living microorganisms! The mineral particles are nonliving, but a teaspoon of healthy soil contains more living things than there are people on Earth.

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🐝 Plants Need Partners

Grade 2 Life Science, Unit 2 — Lesson 2

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Ecosystems 🌱 **Unit:** 2-LS2 Ecosystems

Without pollinators, most of our fruits and vegetables would not exist!

📋 **NGSS Standard:**
2-LS2-2 — Develop a simple model that mimics the function of an animal in dispersing seeds or pollinating plants.

🎯 Learning Objectives

- Understand how pollination works and why it matters for plants
- Identify animals that act as pollinators and seed dispersers
- Build and use a simple model to show how pollen moves between flowers

📦 Materials

🏠 What You Need

- Small bowl of flour (pretend pollen)
- 3-4 drops of yellow food coloring (mix into the flour)
- 1 cotton ball (the "pollinator")
- 2 pieces of paper with a flower drawn on each (or 2 real flowers)
- Paper and pencil for worksheet

✅ *All household items — no specialty purchases needed!*

📖 Vocabulary

Pollination
Moving pollen from one flower to another so the plant can make seeds

Pollen
A tiny yellow powder made by flowers that is needed to make seeds

Pollinator
An animal (like a bee or butterfly) that carries pollen between flowers

Seed Dispersal
Moving seeds away from the parent plant so they can grow in new places

Nectar
A sweet liquid made by flowers that attracts pollinators

Fruit
The part of a plant that holds seeds and often helps spread them

🌍 Background

Plants cannot walk around and find partners on their own. They depend on animals (and wind and water) to help them reproduce. Pollinators like bees, butterflies, and hummingbirds visit flowers to drink sweet nectar. While they feed, tiny pollen grains stick to their bodies — and get carried to the next flower.

When the pollinator flies to the next flower, some pollen rubs off, and that flower can now make seeds. This partnership is so important that 1 in every 3 bites of food you eat exists because of pollinators. Without them, most fruits, vegetables, and nuts would disappear from our world.

Plants also need their seeds to travel. Some seeds travel by wind (dandelions), some by water (coconuts), and some by hitching a ride on animals — in their fur, feathers, or even inside their digestive systems after being eaten in a fruit. Every apple you eat contains seeds that the tree hopes you (or a bird) will carry somewhere new.

🔬 Activity: Pollinator Simulation

In this activity, you will build a simple model of pollination using flour as pollen and a cotton ball as a bee.

💡 **Before you begin:** Mix 3-4 drops of yellow food coloring into a small bowl of flour. Stir until the flour looks yellow. This is your pollen!

Step 1: Draw a flower on each of 2 pieces of paper. Label them Flower 1 and Flower 2. Place them a few feet apart on a table or on the floor.

Step 2: Gently dip the cotton ball into the yellow flour mixture. Look at the cotton ball — the yellow powder clings to it just like pollen clings to a bee's fuzzy body.

Step 3: Touch your "pollinator" cotton ball to Flower 1. Press gently and lift. You have just visited the first flower for nectar!

Step 4: Now "fly" your cotton ball pollinator to Flower 2. Touch it gently to Flower 2 and press. Lift it up. Look at Flower 2 — can you see yellow pollen transferred to it?

Step 5: That yellow flour on Flower 2 is pollination in action! You just modeled how a bee moves pollen from one plant to another, allowing the plant to make seeds.

Step 6: Discuss together: What animal does the cotton ball represent? What does the flour represent? Why do bees visit flowers?

💬 Discussion Questions

🗣️ **Question 1:** What would happen to fruit and vegetables if there were no bees or butterflies?

Suggested answer: Many fruits and vegetables would disappear because they need pollinators to reproduce. Apples, strawberries, cucumbers, almonds, and blueberries all depend on pollinators. Farmers actually rent beehives and bring them to orchards to make sure there are enough bees. Without pollinators, much of our food supply would collapse — roughly 1 in 3 bites of food depends on them.

🗣️ **Question 2:** What other animals besides bees are pollinators?

Suggested answer: Many animals pollinate! Butterflies, moths (at night), hummingbirds, some bats (they pollinate tropical flowers at night), beetles, wasps, and even some flies can all carry pollen. In the rainforest, some plants are only pollinated by one specific species — so if that species disappears, the plant does too.

🗣️ **Question 3:** Why do flowers make nectar and smell good? Who are they "trying to" attract?

Suggested answer: Flowers make nectar and scents to attract pollinators! The bright colors say "look over here!" The scent says "food is here!" The nectar is the reward for visiting. Flowers and pollinators evolved together over millions of years, each benefiting from the other. Scientists call this mutualism — both partners benefit from the relationship.

🐝 Student Worksheet — Plants Need Partners

Name: _____ Date: _____

Part 1 — Pollination Sequence

Draw a picture in each box to show the steps of pollination.

1	2	3	4
Bee visits Flower 1	Pollen sticks to bee	Bee flies to Flower 2	Pollen transfers — seeds can grow!

Part 2 — Draw Your Simulation

Draw your pollinator experiment below. Label: cotton ball, pollen, Flower 1, and Flower 2.

✏️ *Draw here...*

Part 3 — Fill in the Blanks

The _____ carries pollen from flower to flower.

Without pollinators, we would not have _____ .

Part 4 — Circle the Pollinators!

Circle all the animals that can pollinate flowers:

🐝 Bee 🪨 Rock 🦋 Butterfly 🐦 Hummingbird 🚗 Car 🦟 Moth 🦇 Bat

✏️ Pencil

👩‍🏫 Parent/Teacher Notes

- **Materials:** Flour plus food coloring plus a cotton ball. If you do not have food coloring, plain flour still works — show kids how the powder clings to the cotton ball fibers.
- **For a more dramatic effect:** Use two different colors of food coloring for two different "flower species." This shows cross-pollination and species specificity more vividly.
- **Real-world connection:** If you have any flowering plants in your yard, watch for actual pollinators during the activity! Even in cities, bees and butterflies visit any flowering plant they can find.
- **Seed dispersal extension:** After the activity, go outside and find a dandelion that has gone to seed. Blow it gently. Each tiny seed has its own parachute. Ask: why does a dandelion make so many seeds? (Because most will not survive — making many seeds increases the odds.)
- **NGSS connection:** This lesson directly addresses 2-LS2-2: developing a simple model of pollination. The cotton ball simulation IS that model. Ask your student to explain how the model is similar to real pollination, and how it is different.
- **Worksheet answer (circle the pollinators):** Bee, butterfly, hummingbird, moth, and bat are all pollinators. Rock, car, and pencil are not living and cannot pollinate.

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⛓️ Food Chains: Who Eats Who?

Grade 2 Life Science, Unit 2 — Lesson 3

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Ecosystems 🌱 **Unit:** 2-LS2 Ecosystems

Energy flows from the sun, through plants, through animals — and back to the soil.

📋 **NGSS Standard:**
2-LS2-1 — Make observations of plants and animals to compare the diversity of life in different habitats.

🎯 Learning Objectives

- Build and explain a food chain from sun to producer to consumers
- Identify producers, consumers, and decomposers in an ecosystem
- Understand that energy flows through an ecosystem and returns to the soil

📦 Materials

🏠 What You Need

- 6 strips of paper (about 1 inch wide and 5 inches long)
- Markers or crayons
- Tape or a stapler (to link the chain rings)
- About 2 feet of string or yarn (to hang the mobile)

✅ *All household items — no specialty purchases needed!*

📖 Vocabulary

Food Chain
A line showing what eats what, from plants to animals

Producer
A plant that makes its own food from sunlight — it "produces" energy

Consumer
An animal that eats other living things to get energy

Decomposer
A living thing (like a worm or fungus) that breaks down dead plants and animals

Herbivore
An animal that only eats plants

Carnivore
An animal that eats other animals

🌍 Background

Every animal in an ecosystem needs energy to live. That energy originally comes from the sun. Plants capture sunlight and turn it into food through photosynthesis — this makes plants "producers" because they produce energy from sunlight. Animals that eat plants are called herbivores, and they get the plant's stored energy when they eat.

A food chain shows how energy flows from one living thing to the next. A simple food chain might look like this: Sun gives energy to Grass. Grass feeds the Grasshopper. Grasshopper feeds the Frog. Frog feeds the Hawk. Every link in the chain matters — remove one and the whole chain is affected.

When plants and animals die, decomposers like earthworms, fungi, and bacteria break them down. They return nutrients to the soil, which helps new plants grow. Decomposers are the cleanup crew of the ecosystem. Without them, dead things would pile up and plants would run out of nutrients. They complete the cycle and keep the ecosystem healthy.

☀️ Sun → 🌿 Grass → 🦗 Grasshopper → 🐸 Frog → 🦅 Hawk → 🪱 Worm (decomposes)

Energy source | Producer | Herbivore / Consumer | Carnivore / Consumer | Carnivore / Consumer

🔬 Activity: Food Chain Mobile

Step 1: Cut or tear 6 strips of paper (about 1 inch wide and 5 inches long).

Step 2: On each strip, write and draw one of these: SUN ☀️, GRASS 🌿, GRASSHOPPER 🦗, FROG 🐸, HAWK 🦅, WORM 🪱

Step 3: Shape each strip into a loop and connect them like a paper chain, in this order from top to bottom: SUN — GRASS — GRASSHOPPER — FROG — HAWK — WORM.

Step 4: Between each link, you can write a tiny label: "gives energy to" or "is eaten by" — or just let the chain show the order.

Step 5: Punch or cut a small hole at the top of the SUN strip and tie a string through it. Hang your Food Chain Mobile from a doorway or window!

Step 6: Look at your finished mobile and discuss: Where does the energy START? (The sun.) Where does it END? (Back in the soil, thanks to the worm!) Now try removing one link — what happens to all the other links?

💬 Discussion Questions

🗣️ **Question 1:** What would happen to the food chain if all the grasshoppers suddenly disappeared?

Suggested answer: The frogs would lose their main food source and their population would shrink. With fewer frogs, the hawk population would also shrink from lack of food. Meanwhile, the grass would grow unchecked with no grasshoppers eating it — it might take over. Scientists call this a "trophic cascade." Removing one species affects every other species in the chain, both above and below it.

🗣️ **Question 2:** Why are decomposers important even though we rarely notice them?

Suggested answer: Decomposers are the most important recyclers on Earth. They break down dead leaves, dead animals, and waste, releasing nutrients back into the soil. Without them, nutrients would stay locked inside dead things and plants would not be able to grow. Earthworms, fungi (mushrooms), and bacteria are all decomposers. They are working constantly — just underground and out of sight.

🗣️ **Question 3:** The food chain starts with the sun. What would happen to all food chains if the sun disappeared?

Suggested answer: All food chains would collapse. Without sunlight, plants could not photosynthesize and would die. Without plants, herbivores would die. Without herbivores, carnivores would die. Every food chain on Earth starts with the sun — it is the ultimate source of all energy for life on our planet.

⛓️ Student Worksheet — Food Chains

Name: _____ Date: _____

Part 1 — Label the Food Chain

Write the name of each living thing in the box. Then write "Producer," "Herbivore," or "Carnivore" on the line below each animal.

☀️ Sun	🌿 Grass	🦗 Grasshopper	🐸 Frog	🦅 Hawk
Energy source				

Part 2 — Fill in the Blanks

A _____ makes its own food from sunlight.

A _____ eats only plants.

A _____ eats other animals.

A _____ breaks down dead things and returns nutrients to the soil.

Part 3 — Draw Your Own Food Chain

Think of animals in your backyard or neighborhood. Draw a food chain with 4 links. Include arrows showing the direction energy flows.

→ → →

Part 4 — Think About It

If I removed the FROG from the food chain, I think this would happen:

_____

_____

👩‍🏫 Parent/Teacher Notes

- **The paper chain mobile is the model** called for in this lesson. Hang it from a doorway or window where students can see it all week as a visual reference.
- **Key vocabulary to focus on:** producer, consumer, decomposer. If students remember just these three words, they understand the structure of every food chain on Earth.
- **Advanced learners:** Introduce the idea of a food WEB — multiple food chains overlapping. A frog might eat grasshoppers AND flies AND worms. Draw connecting arrows to show the web and how it is more resilient than a simple chain.
- **Decomposer real-world connection:** If you have any soil at home, dig a small hole and look carefully. You will likely find earthworms. Hold one up and celebrate — that is a decomposer at work! Students are often delighted to meet one in real life.
- **Connection to Lesson 6:** Students will use food chain knowledge when they design their shoebox habitat in the unit culminating project.
- **Common misconception:** Students sometimes think the energy is "used up" as it moves through the chain. Actually, energy is transferred — each animal gets a portion of the previous animal's stored energy. Some is lost as heat at each step, which is why longer food chains have fewer top predators.

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🦊 Predator and Prey: The Population Game

Grade 2 Life Science, Unit 2 — Lesson 4

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 minutes 🔬 **Subject:** Life Science — Ecosystems 🌱 **Unit:** 2-LS2 Ecosystems

Predators and prey keep each other's populations in balance.

📋 **NGSS Standard:**
2-LS2-1 — Make observations of plants and animals to compare the diversity of life in different habitats.

🎯 Learning Objectives

- Understand the relationship between predators and prey
- Observe how predator and prey populations affect each other over time
- Recognize that these relationships help keep ecosystems in balance

📦 Materials

🏠 What You Need

- 10 small red paper scraps (or any colored cards) as "life cards"
- Open space: living room, hallway, or yard
- Pencil and tally sheet (Student Worksheet)

✅ *No specialty purchases needed — just paper scraps and open space!*

📖 Vocabulary

Predator
An animal that hunts and eats other animals

Prey
An animal that is hunted and eaten by another animal

Population
The total number of one kind of animal living in an area

Balance
When predator and prey populations stay at healthy levels over time

Reproduce
To have babies and make more of your species

Survive
To stay alive despite challenges from predators or the environment

🌍 Background

In every ecosystem, some animals eat other animals. The animal that hunts is called the predator. The animal being hunted is called the prey. Foxes hunt rabbits. Hawks hunt mice. Owls hunt voles. These relationships keep ecosystems healthy by preventing any one species from becoming too numerous.

Predator and prey populations are connected like a seesaw. When there are lots of rabbits (prey), foxes (predators) have plenty to eat and their population grows. But more foxes eat more rabbits, so rabbit numbers start to drop. With fewer rabbits, some foxes go hungry and fox numbers drop too. With fewer foxes, rabbits bounce back — and the cycle starts again.

This up-and-down cycle keeps ecosystems in balance. It is not cruel — it is the way ecosystems stay healthy. When humans remove predators from an ecosystem (such as by hunting wolves away from a forest), prey populations can explode, eat all the vegetation, and cause the ecosystem to collapse. Every predator plays a role, even the scary ones.

🔬 Activity: Population Tag

💡 **Setup:** 1 player = fox (predator). All other players = rabbits (prey). Each rabbit starts with 2 red cards ("lives") held in their hands. This works great with just 2 people (one adult as fox, one child as rabbit) or a whole family!

Round Setup: Fox stands at one end of the room. Rabbits scatter in the space. Give each rabbit 2 red cards to hold.

Play (1 minute each round): Fox chases rabbits and tags them. Each tag = fox takes 1 life card. A rabbit with 0 cards is "eaten" and sits down for that round.

Count after each round: How many rabbits survived? How many were eaten? Write the numbers in your tally sheet.

Reproduce: Each surviving rabbit gets 1 new card (they had a baby rabbit!). Fox stays in the game if they tagged at least 1 rabbit.

Repeat for 5 rounds total. After Round 5, look at your numbers. Did the rabbit population go up and down in a pattern?

💬 Discussion Questions

🗣️ **Question 1:** What happened to the rabbit population over 5 rounds? Did it stay the same, or did it change?

Suggested answer: Usually the population zigzags! When rabbits are plentiful, the fox catches many and rabbit numbers drop. When rabbit numbers drop, the fox struggles to catch enough — then rabbits recover. This is the natural predator-prey cycle. Even in a short 5-round game, students often see this balance develop.

🗣️ **Question 2:** What would happen if we added 3 more foxes to the game?

Suggested answer: More foxes would eat more rabbits faster, driving the rabbit population down quickly. With very few rabbits left, the many foxes would struggle to find food and some would "starve." Eventually the system might balance at a lower level — or collapse if the rabbits go extinct. This is why introducing or removing predators from an ecosystem has huge ripple effects.

🗣️ **Question 3:** If all the rabbits disappeared from an ecosystem, what would happen to the foxes?

Suggested answer: The foxes would lose their food source. Some might find other prey (mice, birds), but if rabbits were their main food, the fox population would shrink significantly. This shows that predators NEED their prey to survive — they are just as dependent on each other as plants and pollinators.

🦊 Student Worksheet — Predator and Prey

Name: _____ Date: _____

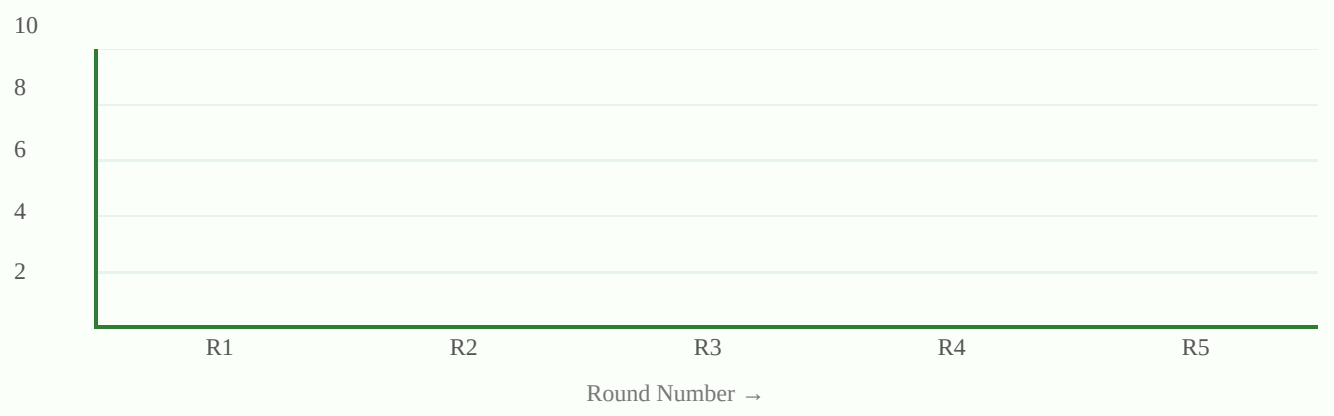
Part 1 — Population Tally Table

Record your data after each round of Population Tag.

Round	🐇 Rabbits at Start	🦊 Rabbits Eaten	🐇 Rabbits Survive	🐇 After Reproduce
1				
2				
3				
4				
5				

Part 2 — Graph the Rabbit Population

Plot your rabbit "After Reproduce" number for each round. Connect the dots with a line. What shape does it make?



Part 3 — Vocabulary Match

Draw a line from each word to its definition.

Predator	An animal that is hunted
Prey	When populations stay at healthy levels

Population	The total number of one kind of animal in an area
Balance	An animal that hunts other animals

Part 4 — Reflection

I noticed that when the fox population goes up, the rabbit population goes _____ .

This happens because _____ .

👩‍🏫 Parent/Teacher Notes

- **Works great with just 2 people:** One adult as fox, one child as rabbit (or vice versa). With more siblings, add more rabbits and possibly 2 foxes.
- **No red cards?** Use any 2 different colored paper scraps, or just have rabbits hold up 2 fingers as "lives." The mechanic is the same.
- **The point is not who wins.** The point is to notice the population numbers going up and down. Stop after each round to count and record — data collection IS the lesson.
- **Real-world connection — Yellowstone wolves:** When wolves were removed from Yellowstone in the 1920s, elk populations exploded and ate all the riverside vegetation. When wolves were reintroduced in 1995, elk moved around more, vegetation recovered, rivers changed course, and beaver populations returned. This "trophic cascade" is one of the most famous ecosystem stories in science. Share it if your student is captivated!
- **Connection to Lesson 3:** Predator-prey relationships ARE food chains in action. Connect back to the food chain mobile: the fox and rabbit are two links in a larger chain.

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🔍 Habitat Hunt: Explorer's Field Journal

Grade 2 Life Science, Unit 2 — Lesson 5

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 45 min (20 outdoor + 25 journaling) 🔬 **Subject:** Life Science — Ecosystems
🌱 **Unit:** 2-LS2 Ecosystems

Scientists observe carefully and write down everything they notice. Today, YOU are the scientist!

📋 **NGSS Standard:**
2-LS2-1 — Make observations of plants and animals to compare the diversity of life in different habitats.

🎯 Learning Objectives

- Observe and record living and nonliving things in a local habitat
- Practice careful scientific observation and field journaling
- Connect real-world findings to ecosystem concepts from earlier lessons

📦 Materials

🏠 What You Need

- Clipboard or notebook (or this printed worksheet as your field journal)
- Pencil
- Outdoor access (yard, park, sidewalk, or any outdoor space)
- Magnifying glass — optional but highly recommended!

✅ *All household items — just a pencil and some outdoor time!*

📖 Vocabulary

Observation
Looking carefully and describing exactly what you see, hear, or touch

Habitat
The specific place where an animal or plant lives and finds what it needs

Species
One kind of living thing (all dogs are one species; all robins are one species)

Diversity
Having many different kinds of living things in one place

Micro-habitat
A tiny habitat inside a bigger one (example: under a rock is a micro-habitat)

Field Journal
A notebook where scientists write down what they observe in nature

🌍 Background

Scientists who study ecosystems spend a lot of time outdoors, looking carefully and writing down what they observe. This careful looking is called observation. A good field journal records what you see, where you saw it, the time, and the weather conditions. Even a small backyard or a crack in the sidewalk is full of living things waiting to be discovered.

Every habitat has a mix of living and nonliving things. Even a patch of concrete has nonliving things (concrete, sunlight, shadow) and living things (ants, weeds growing through cracks, maybe a bird landing for a moment). The richer the habitat — the more plants, soil types, and moisture it has — the more diverse the life you will find.

Scientists discover new species every year, even in places humans have lived for centuries. An acre of healthy soil contains billions of living things too small to see. One square foot of leaf litter can hold hundreds of insects and microorganisms. The world is full of life — you just have to slow down and look carefully.

🔬 Activity: 20-Minute Habitat Hunt

💡 **Before you go:** Complete the "Predictions" section of your Student Worksheet. Predicting first helps you notice what was surprising when you come back inside.

Step 1: Head outside with your field journal (Student Worksheet) and pencil. Set a timer for 20 minutes.

Step 2: Go slowly. Look under rocks, in grass, on leaves, in soil, and around plants. The slower you move, the more you find.

Step 3: Record every living thing you spot in your observation log. Write its name if you know it, or describe it: "small brown beetle with spots," "gray spider near web on fence."

Step 4: Record nonliving things too — types of soil, rocks, water sources (puddles, streams), sunlight, shadows.

Step 5: Choose 2 things to sketch in detail. Take your time. Label the parts you notice (legs, wings, color, size, where found).

Step 6: Come back inside and count: how many total living things did you find? How many different types (species)? Compare your actual findings to your predictions — what surprised you?

💬 Discussion Questions

🗣 **Question 1:** What did you find the most of? Was that what you expected?

Suggested answer: Most students find insects (especially ants) are the most numerous, followed by plants. Most people are surprised by how much life is in a tiny area once they slow down and look carefully. What is common varies by habitat — a lawn has lots of grass and ants; a garden has more variety; a weedy area near a fence might have the most diversity of all.

🗣 **Question 2:** Did you find any micro-habitats — tiny habitats within the bigger habitat?

Suggested answer: Under a rock, under a log, in a crack in the sidewalk, inside a flower, or under a leaf — these are all micro-habitats. Each one has slightly different conditions (darker, moister, more protected) that suit different organisms. Scientists call these "microenvironments." Even a single flower is a habitat for tiny insects and spiders.

🗣 **Question 3:** What was the most surprising living thing you found? What do you think it needs to survive?

Suggested answer: Guide students to think about what the organism needs: food, water, shelter, and the right temperature. Then ask: does this place provide those things? Why is this animal or plant here and not somewhere else? Every organism is in its habitat for a reason — because that specific place meets all of its needs.

🔍 Field Journal — Habitat Hunt

Name: _____ Date: _____ Time: _____ Weather: _____

Part 1 — My Predictions (Before I Go Outside)

Write 5 living things and 3 nonliving things you think you will find:

Living things I predict:

_____

_____

_____

_____

_____

Nonliving things I predict:

_____

_____

_____

Part 2 — Observation Log (While Outside)

🌿 Living Thing Found	📍 Where I Found It

Nonliving things found:

_____

_____

_____

🎨 Detailed Sketches

Name: _____

Choose 2 things you found and sketch them carefully. Label the parts you notice.

Sketch 1

Name:

Where I found it:

Sketch 2

Name:

Where I found it:

Part 3 — Count and Reflect

Total living things I found: _____ Total different types (species): _____

The most surprising thing I found was _____
because _____ .

One thing I predicted that I DID find: _____
One thing I did NOT expect to find: _____ .

👩‍🏫 Parent/Teacher Notes

- **Any outdoor access works:** A city sidewalk, a balcony with a potted plant, a park, or a driveway edge — anywhere living things can be found. If outdoor access is truly impossible, create an indoor "habitat hunt" using houseplants or a cup of soil from outside brought in to examine.
- **Bring a magnifying glass:** Even a cheap plastic one transforms this activity. The world under a magnifying glass is endlessly fascinating for children. Under 10x magnification, an ant becomes an alien, a leaf surface becomes a landscape.
- **Best timing:** Morning hours, when insects are active. Even 15 minutes outside is enough for this lesson.
- **Taxonomy tip:** Students will find things they cannot name. That is great! Scientists encounter unnamed things all the time. Describe the features carefully and look it up later using a field guide app like iNaturalist (free, works offline, identifies from photos).
- **Photography extension:** If your student has access to a tablet or phone camera, let them photograph their 2 detailed sketch subjects. This creates a digital field journal and often motivates more careful observation.
- **Connection to Lesson 1:** Ask students to connect what they found to the Ecosystem Map from Lesson 1. Did they find more living or nonliving things? What nonliving things did the living things depend on?

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🏗️ Design a Habitat: Build a Home for a Creature

Grade 2 Life Science, Unit 2 — Lesson 6 (Unit Culminating Project)

📚 **Grade:** 2nd Grade ⏱️ **Duration:** 60+ minutes 🔬 **Subject:** Life Science — Ecosystems
🌱 **Unit:** 2-LS2 Ecosystems — Culminating Project

Every animal needs the right food, water, shelter, and space — can you build it?

📋 **NGSS Standard:**
2-LS2-1 + **K-2-ETS1-1** — Make observations of plants and animals to compare the diversity of life in different habitats AND ask questions, make observations, and gather information to define a simple problem and design a solution.

🎯 Learning Objectives

- Engineer a model habitat that meets the needs of a chosen animal
- Apply ecosystem knowledge from the whole unit to a real design challenge
- Explain the purpose of each element in a habitat model

📦 Materials

🏠 What You Need

- A shoebox or small cardboard box (the habitat container)
- Soil or dirt from outside
- Pebbles or small rocks
- Grass clippings or dry leaves
- A few small sticks or twigs
- A bottle cap with a little water (for a water source)
- Construction paper and markers (for labels and decorating)

✅ *All household or backyard materials — no specialty purchases!*

📖 Vocabulary

Engineer
A person who designs and builds things to solve problems

Model
A small version of something real, used to study and test ideas

Need
Something a living thing must have to survive (food, water, shelter, space)

Shelter
A safe place that protects an animal from weather and predators

Design
A plan for how to build or make something to solve a problem

Criteria
The requirements a design must meet to be considered successful

🌍 Background

Every animal has a habitat that meets its needs. A habitat must provide food, clean water, shelter from predators and weather, and enough space. When scientists want to protect an endangered species, they first study its natural habitat and then try to create or preserve places that meet all those same needs.

Engineers design things to solve problems. When you design a habitat model, you are acting as both a scientist (understanding the animal's needs) and an engineer (building something to meet those needs). This kind of thinking — identifying needs, designing solutions, and testing and improving — is called the Engineering Design Process.

Today you will choose one small animal: an earthworm, a ladybug, or a snail. You will find out what it needs, then build a model habitat from everyday household materials. By the end, you will be able to explain why every part of your habitat is there and what animal need it meets.

🐾 Know Your Animal

Read the animal info card for your chosen animal before you start designing:

🪱

EARTHWORM

- **Food:** Dead leaves, decaying plant matter, soil particles
- **Water:** Moist (not soaking wet) soil
- **Shelter:** Underground tunnels in dark soil
- **Space:** A few square inches per worm
- **Build tip:** Damp soil + torn dry leaves on top for food

🐞

LADYBUG

- **Food:** Tiny aphids (plant pests) and some pollen
- **Water:** Dew or tiny water droplet
- **Shelter:** Under a leaf or bark, protected from rain
- **Space:** Very small — a few square inches
- **Build tip:** Leaf "roof" + small sticks to climb + grass for aphid habitat

🐌

SNAIL

- **Food:** Soft plant matter, algae, tender leaves
- **Water:** Moist environment — dries out in sun
- **Shelter:** Shady, cool corner (carries its own shell)
- **Space:** Small — but needs room to move
- **Build tip:** Mist with water + large leaves for shade and food + bottle-cap pond

🔬 Activity: Design and Build Your Habitat

Step 1 — Choose: Pick your animal: earthworm, ladybug, or snail. Write its name on your Student Worksheet.

Step 2 — Research: Read the animal info card above. Fill in the needs chart on your Student Worksheet: what does your animal need for food, water, shelter, and space?

Step 3 — Plan: On the design sketch page in your Student Worksheet, draw a top-down view of your habitat and label what you plan to include and where.

Step 4 — Build: Gather your materials and build the habitat inside the shoebox. Add each element one at a time. As you add each piece, ask: "Which need does this meet: food, water, shelter, or space?"

Step 5 — Label: Make small flag labels (paper folded over a stick or pencil) for every part of your habitat. Each label should say what the element is AND which animal need it meets.

Step 6 — Present: Explain your habitat to a family member. Tell them: what animal lives there, what each part provides, and how you know your animal could survive there.

💬 Discussion Questions

🗣️ **Question 1:** What would happen to your animal if one part of the habitat was removed? Pick one part and explain.

Suggested answer: Every part of the habitat serves a need. If you removed the water source, the snail or earthworm would dry out and die. If you removed the leaves, the earthworm would have nothing to eat. If you removed the shelter (the leaf roof), the ladybug would be exposed to predators and weather. Every element is there for a reason — this shows how fragile real habitats are when even one thing changes.

🗣️ **Question 2:** How is your model habitat the same as a real habitat? How is it different?

Suggested answer: The same: it provides food, water, shelter, and space. Different: it is much smaller, it does not have weather or seasons, it lacks predators and the community of other animals the real animal would interact with. A model is always a simplified version used to test ideas. Scientists use models all the time and always note their limitations — a model helps us understand without being as complex as reality.

🗣️ **Question 3:** If you could add one more thing to improve your habitat, what would it be and why?

Suggested answer: This is open-ended by design! Students might add more food, a better shelter, other species to make it more realistic, or more space. The key is that students justify their choice by linking it to a specific animal need. This is engineering iteration — engineers are always looking for ways to improve a design after testing it.

🏗️ Student Worksheet — Design a Habitat

Name: _____ Date: _____

Part 1 — My Animal

I chose the: [ ] 🪱 Earthworm [ ] 🐞 Ladybug [ ] 🐌 Snail

Part 2 — Animal Needs Chart

Need	What my animal needs
🍃 Food	
💧 Water	
🏠 Shelter	
📐 Space	

Part 3 — My Habitat Design Sketch

Draw a top-down view of your habitat plan. Label what you will add and which need it meets.

✏️ *Draw your habitat plan here. Label everything!*

🏗️ Student Worksheet — Design a Habitat (continued)

Name: _____

Part 4 — Materials List

Material I Used	Which Need It Meets

Part 5 — Habitat Checklist

After building, check off each item:

- ☐ My habitat has a food source for my animal
- ☐ My habitat has a water source
- ☐ My habitat has shelter (protection from weather or predators)
- ☐ My habitat has enough space for my animal to move
- ☐ Every part of my habitat is labeled with what need it meets

Part 6 — Reflection

If I were a _____ , I could survive in my habitat because _____ .

One thing I would add or change to make my habitat better:

_____

👩‍🏫 Parent/Teacher Notes

- **This is the UNIT CULMINATING PROJECT.** Give it 60+ minutes and let the student truly own it. Ownership of the animal choice dramatically increases engagement.
- **Let them choose:** Earthworm for a child who likes dirt and digging; ladybug for a child interested in insects; snail for a child who likes careful, slow observation. All three options work with backyard materials.
- **If you have the real animal nearby** (earthworms in garden soil, ladybugs on plants), consider letting the student observe the real animal before building its model habitat. This connects the model to reality in a powerful way.
- **Engineering Design Process:** This lesson hits K-2-ETS1-1 directly. Students define the problem (what does the animal need), design a solution (plan and build the habitat), and evaluate it (does it meet the criteria checklist). Consider doing a brief "test" — if you can find the actual animal, place it in the habitat briefly and observe its behavior.
- **Differentiation:** For advanced students, have them research a more complex animal (a specific bird or amphibian) and add complexity. For younger siblings, have them help build while the 2nd grader plans and labels.
- **Photography suggestion:** Photograph the finished habitat! It makes a great portfolio piece and motivates students to do their best work. If printing this unit, the photo could be added to a student science portfolio.
- **Unit synthesis connection:** Ask students to name the ecosystem their animal is part of. What food chain includes their animal? What would pollinate the plants in their habitat? This lesson is the synthesis of the entire unit — draw connections back to Lessons 1-5.

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🌿 Living Things Need Each Other

ECOSYSTEMS · FOOD CHAINS · HABITATS · POLLINATORS

🌍

ECOSYSTEMS

- Living + nonliving together
- Plants, animals, soil, water
- All parts are connected
- Change one → change all

⛓️

FOOD CHAINS

- Energy flows from the sun
- Sun → Plant → Herbivore → Carnivore
- Arrows show energy direction
- Remove one link → chain breaks

🐝

🏡

☀️ **NO SUN**
No plants, no food chain

🐝 **NO POLLINATORS**
No fruit or seeds

🪱 **NO DECOMPOSERS**
Dead things pile up

💧 **NO CLEAN WATER**
Ecosystem collapses

🐛 One earthworm can eat half its body weight in soil daily

🌻 A single oak tree supports 500+ species

🦋 Monarch butterflies migrate 3,000 miles

🐝 Bees can recognize human faces

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🌿 Ecosystems Quiz

Unit 2 — Life Science • 2nd Grade

📖 **Ask a grown-up to read the questions to you!**

⭐ **Score: 0 / 10**

🌳 **1** **What is an ecosystem made of?**

- Only living things like plants and animals
- Living AND nonliving things working together
- Only water and soil
- Only animals that eat each other

🌿 **2** **Which of these is a NONLIVING part of an ecosystem?**

- A butterfly
- A mushroom
- A rock
- A worm

🐝 **3** **What does a bee carry from flower to flower that helps plants make seeds?**

- Nectar
- Pollen
- Soil
- Water

☀️ **4** **In a food chain, where does all the energy originally come from?**

- The soil
- The rain
- The sun
- The biggest animal

🌿 **5** **What do we call a living thing that makes its own food from sunlight?**

- Consumer
- Decomposer
- Producer
- Predator

🦊 **6** **A fox hunts and eats rabbits. What is the fox?**

Prey

Predator

Producer

Decomposer

7 **What does a decomposer do in an ecosystem?**

It makes food from sunlight

It hunts other animals

It breaks down dead things and returns nutrients to the soil

It carries pollen between flowers

8 **What 4 things must every animal habitat provide?**

Sun, rain, wind, and soil

Food, water, shelter, and space

Predators, prey, plants, and rocks

Pollen, nectar, seeds, and fruit

9 **Which of these animals is a pollinator?**

An earthworm

A fox

A butterfly

A hawk

10

In the food chain Sun → Grass → Grasshopper → Frog → Hawk, what would happen if the grasshoppers all disappeared?

Nothing would change

The frog and hawk populations would shrink, and grass would grow uncontrolled

The hawk would eat more frogs

The sun would get brighter

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